



**ELSEWEDY
UNIVERSITY
OF TECHNOLOGY**
POLYTECHNIC OF EGYPT

VERSION 0.0



Incident investigation

From First Response to High-Level Analysis



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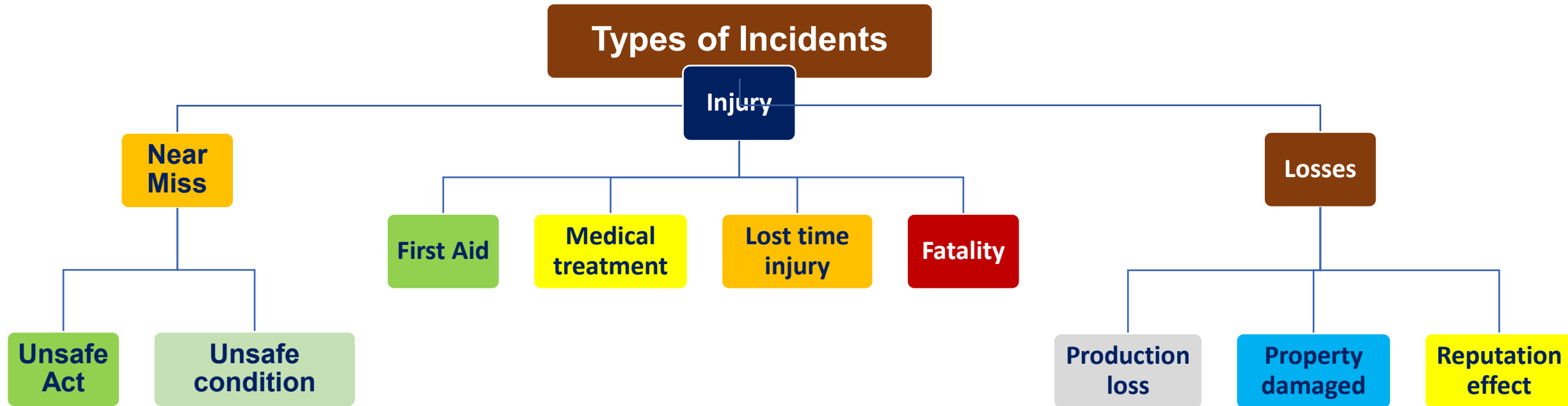
Objectives of the Session

1. Understand the definition and importance of incident investigation
2. Learn the immediate steps to take after an incident
3. Explore data collection and evidence-gathering techniques
4. Master root cause analysis tools (5 Whys, Fishbone, etc.)
5. Develop practical corrective and preventive actions
6. Learn how to document and communicate findings effectively
7. Participate in a real-life investigation simulation

What is an Incident?

- Unplanned and undesired event
- Could have resulted in:
 - Injury or illness
 - Damage to property
 - Environmental harm
 - Business interruption

Types of Incidents



Why Do We Investigate Incidents?

- To prevent recurrence of similar events
- To identify root causes, not blame people
- To comply with legal and insurance requirements
- To improve safety culture and awareness
- To enhance risk control measures



Immediate Actions After an Incident

1. Ensure Safety

1. Provide first aid
2. Evacuate if needed
3. Control immediate hazards

2. Preserve the Scene

1. Don't move equipment unless needed for safety
2. Take photos/videos before disturbing the area

3. Notify Authorities

1. Inform HSE department, supervisors, and emergency response if needed

4. Secure the Area

1. Use barrier tape/signage
2. Prevent unauthorized access

Incident investigation process

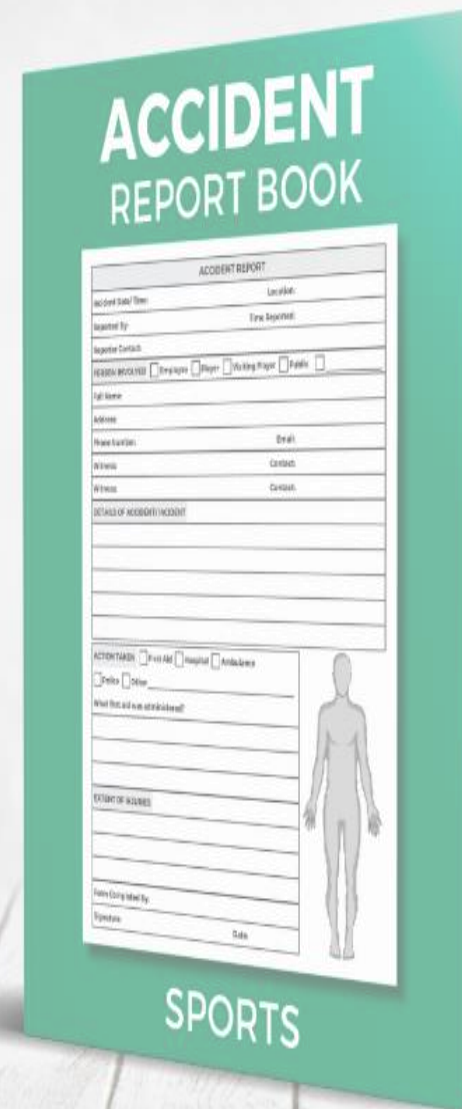


Initial Reporting :

❖ Initial Reporting Requirements Information to Collect Immediately:

- Date and time of incident
- Location
- People involved (names, roles)
- Description of what happened
- Witnesses
- Injuries or damages
- Initial response actions taken

Tool: Incident Notification or Initial Report Form



Data Collection Techniques

1.Observations & Inspections

1. Take clear, timestamped photographs
2. Note hazards, conditions, warning signs

2.Interviews

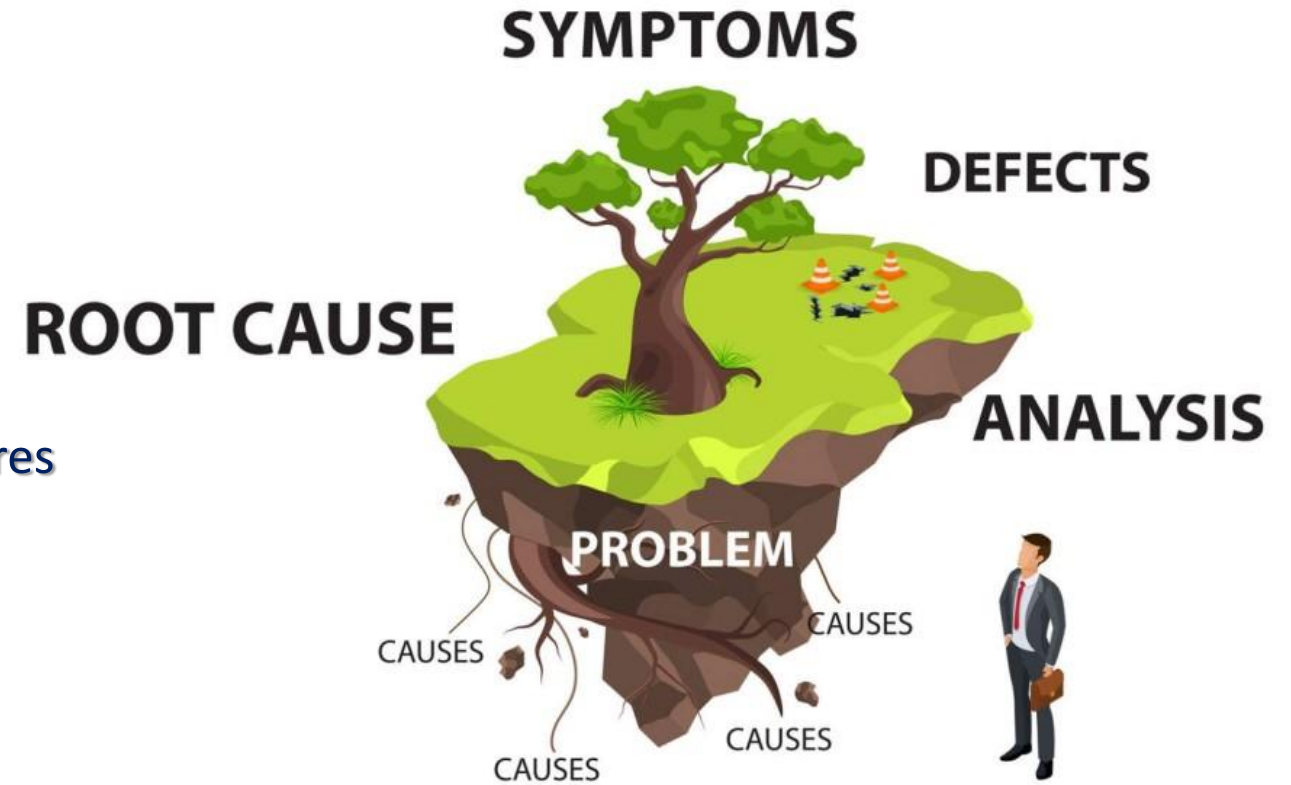
1. Interview involved persons and witnesses
2. Use open-ended questions: "What did you see?" "What were you doing?"

3.Document Review

1. Training records
2. Maintenance logs
3. Risk assessments
4. Permit-to-work systems

Root Cause Analysis – Introduction Levels of Causes:

- Immediate/Direct Cause: Unsafe act or condition
- Underlying/Indirect Cause: System/process issues
- Root Cause: Management, planning, or cultural failures



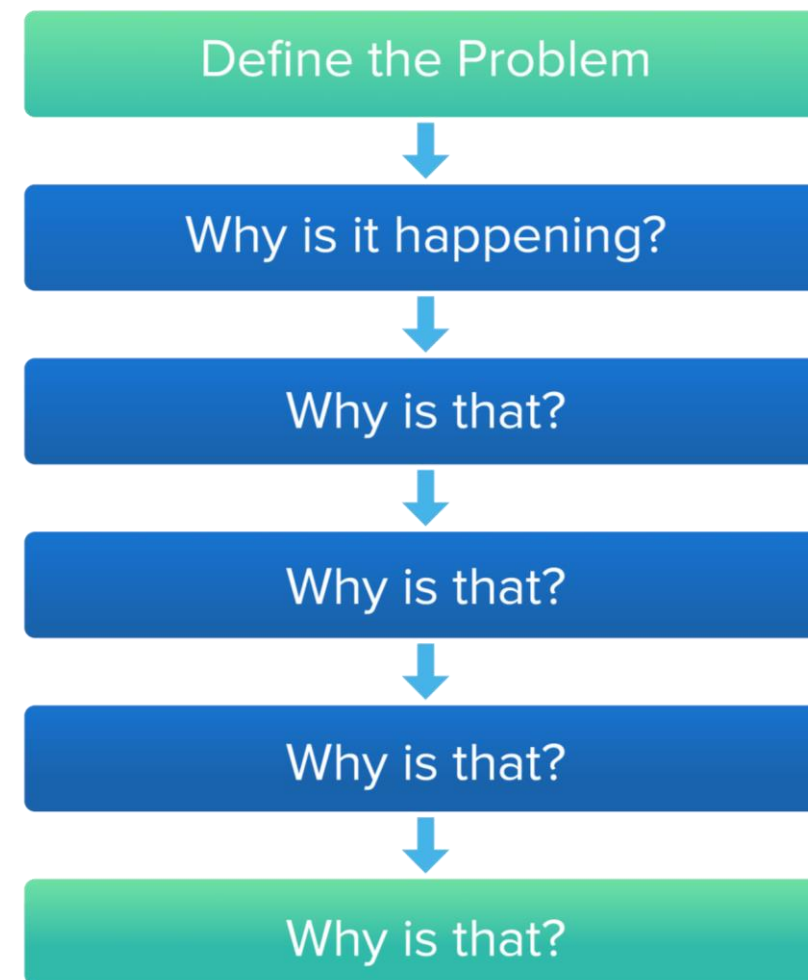
Root Cause Tools – 5 Whys The 5 Whys Technique:

- Ask "Why?" repeatedly until the real cause is identified

Example:

- 1.Why did the worker slip? – Because the floor was wet
- 2.Why was the floor wet? – Because a pipe leaked
- 3.Why did the pipe leak? – Because it wasn't maintained
- 4.Why wasn't it maintained? – No maintenance schedule
- 5.Why is there no schedule? – No preventive maintenance system

The 5 Whys



Root Cause

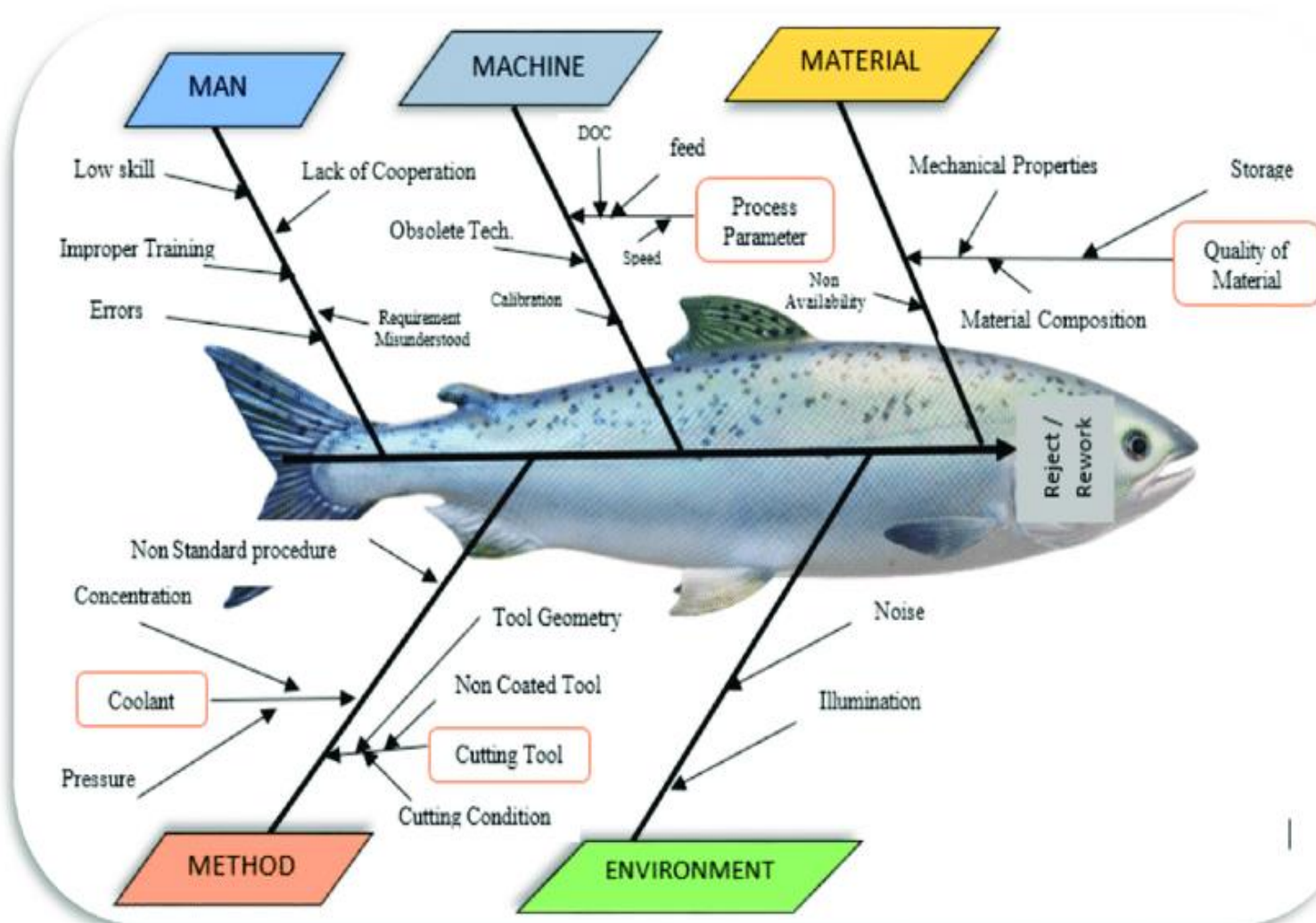
Root Cause Tools – Fishbone Diagram

- Also called **Ishikawa** Diagram
- Organize possible causes under categories

Categories:

- People
- Process
- Equipment
- Environment
- Materials
- Management

Tip: Use this with teams to brainstorm



Case Study – Investigation Practice Scenario

- A student suffers a minor electric shock in a lab

Instructions:

- Identify immediate actions
- Collect evidence
- Conduct 5 Whys or Fishbone Analysis
- Suggest root cause
- Propose corrective and preventive actions



Corrective & Preventive Actions (CAPA)

Corrective Action Actions:

- Fix the issue
- Example: Replace damaged cable

Preventive Actions:

- Prevent recurrence
- Example: Periodic inspections

SMART Actions:

- Specific
- Measurable
- Achievable
- Relevant
- Time-bound



Follow-Up & Effectiveness Review

- Assign responsibilities and deadlines
- Verify action completion
- Conduct re-inspection
- Monitor similar areas
- Update procedures/training



Final Investigation Report Contents:

- Incident summary
- Investigation findings
- Root cause analysis
- Evidence/photos
- CAPA
- Signatures and approvals

Tip: Keep confidential and focus on systems

ELSEWEDY ELECTRIC			
Incident Report			
1 General Information Name: Ahmed Hassan / security officer Supervisor: Ramadan Farag Title: electrical technician Date: 28-12-2024 Location: Elastroid Section / 300-10 compressors ☐ Near Job ☐ Accident ☐ Injury ☐ ADI ☐ Injury ☐ Loss ☐ Fire ☐ equipment damage		Type of activity: Startup for the press Location: Elastroid Section / 300-10 compressors	
2 Description of event On December 28, 2024, during the night shift at 1:45 AM, an electrical technician started up the compressor. Approximately 10 minutes after startup, a fire ignited, and flames were observed. The fire detection system was immediately activated. Two security team members, along with the electrical technician, responded promptly to the incident. They subsequently, the Civil Defense was contacted, and the fire was successfully brought under control by 2:35 AM.			
3 Direct Causes *short circuit of electrical connection of heater		Root Causes *inadequate wiring cause ignition source	
4 Immediate Cause Of Incident:			
5 Job Related Causes: ☐ Inadequate Procedures ☐ Inadequate Tools/Equipment/P.P.E ☐ Inadequate Planning ☐ Inadequate Resources ☐ Inadequate Supervision/Competency ☐ Inadequate Design/Engineering ☐ Inadequate Administrative Supervision ☐ Others (Specify):		Personal Related Causes: ☐ Inadequate Skills/Knowledge ☐ Operating Without Authority ☐ Inadequate Training ☐ Inadequate Communication ☐ Disregard Of Procedures ☐ Unable To Do Job Physically ☐ Inadequate Appreciation Of Situation ☐ Others (Specify):	
6 Evidence/Photos 			
7 Corrective Action – To Describe Actions already Taken:			
Corrective Action		Responsible Person	
*review the preventive maintenance schedule and implementation plan.		MAINTAINANCE	
*include the housekeeping of the entire of the compressor		MAINTAINANCE	
8 Preventive action to be taken		Due Date	
*All machines, whether new or used, must undergo maintenance in accordance with the manufacturer's recommendations. (MAINTAINANCE)		31/3/2025	
9 Signatures and Approvals		Date:	
Investigation Team:			
Name:		Signature:	
Ahmed Hassan		HSE section head	
Mohamed Tawfik		HSE Manager	
Ehab Ahmed		Production director	
Ahmed Abdelaziz		Maintenance manager	
Mohamed Abdelhameed		Product director	
Witnesses			
Name:		Signature:	
Ahmed Hassan		security officer	
Mohamed Ehab		security supervisor	

Legal & Ethical Considerations

- Maintain confidentiality
- Avoid blaming individuals
- Comply with laws and reporting regulations
- Involve authorities if needed



Key Takeaways

- Incidents = learning opportunities
- Focus on "why," not "who"
- Root causes are often systemic
- Use 5 Whys and Fishbone tools
- Always follow up on actions

Questions & Open Discussion

- What was challenging?
- What would you improve?

Group Simulation Assignment Instructions - Quiz

1. Form teams

2. Get incident scenario

3. Investigate (respond, collect, analyze)

4. Propose actions

5. Present findings

Goal: Practice full cycle of investigation





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